BANSEEDHAR GONDALIYA

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PROFESSIONAL SUMMARY

Results-driven **Machine Learning Engineer** with expertise in **Deep Learning, Generative AI, and Agentic AI**. Experienced in building scalable, production-grade solutions across **Natural Language Processing (NLP)**, **Computer Vision**, and **multi-agent orchestration**. Holds multiple **patents** and **peer-reviewed publications**, with a strong focus on translating cutting-edge AI research into real-world impact.

EXPERIENCE

IDFC FIRST Bank

April 2026 – Present

Machine Learning Engineer | *Gen AI Team* — Bengaluru, India

- Developed an **enterprise-grade autonomous agentic AI system** using **LangGraph, DeepAgents, and AWS AgentCore** that trains, tunes, evaluates, explains, and reports **banking fraud detection models** from natural language instructions, orchestrating **task planning, specialized agent delegation, tool execution, memory, and skills** to automate the end-to-end ML lifecycle.
- Building a **query agent** using **LangGraph and AWS AgentCore** to fetch data from **datamarts** (bureau, credit card, loans, etc.) from natural language queries, leveraging **datamart metadata as semantic context** for columns/tables and **Neo4j graph-based code context** within a **multi-step agentic system** for accurate query generation.

Samsung Research and Development

Aug 2023 – Present

Senior Machine Learning Engineer | *AI Productization Team* — Bengaluru, India

- **Generative AI & Agentic AI:**
  - Engineered a **Screen AI Agent** integrating LLMs with **UI understanding** (element detection, icon classification, view hierarchy), reducing task time by 55% and improving user satisfaction; outcomes include an **ICVGIP 2025 paper**, a granted **Samsung AI patent**, and feature release as **Dynamic Interactive Task Guidance**.
  - Fine-tuned Qwen2.5-Coder-32B-Instruct (0.5B parameters) using full **fine-tuning** and **LoRA(Low-Rank Adaptation)** to bridge LLM natural language responses to **dynamic UI representations**, enabling seamless rendering of interactive user interfaces.
  - Developed **Generative AI solutions** for summarization and translation using **LLMs for Bixby's Android integration**, enhancing contextual interaction and language understanding.
- **Natural Language Processing(NLP) & Data Science:**
  - Built an **event classification model** for user interaction commands (address, email, event, phone, URL, etc.) using **ReTVec, TinyBERT, and LSTM**, achieving 96% accuracy; commercialized in the **ODCR module** with support for 9 languages.
  - Developed a **relationship prediction model** using conversational and behavioral data (calls, messages), leveraging **LSTM, BiLSTM, BERT**, achieving 95% accuracy across 7 classes; innovation resulted in a **granted patent**.
  - Designed and implemented an **OCR text denoising** solution using **encoder-decoder** architectures **(RNN, Transformer)**, significantly improving text recognition accuracy by **91%**.
- **Computer Vision (CV):**
  - Contributed to the development of **Screen AI and AI Select** technologies using on-screen **object detection** (e.g., timestamps, keyboards, maps) with YOLO, achieving an inference time of **150ms** and reducing the model size from **20MB to 1.5MB**.
  - Integrated the optimized UI element detection model into **On-Device Context Retriever (ODCR)** module to reduce textual noise and enhance **entity extraction** like phone numbers, emails, addresses, URLs, etc., enabling smart suggestions.

Samsung Research and Development

May 2022 - July 2022

Software Developer Engineer - Intern | *OnDevice AI Team* — Bengaluru, India

- Worked on designing and implementing a versatile **Machine Learning model for Human Activity Recognition (HAR)** using sensor data from wearable devices.
- Conducted extensive research and evaluation of multiple HAR methodologies to ensure high functionality and accuracy standards.
- Developed a **Deep Learning-based solution** leveraging a **Dimension Adaptive Neural Architecture**, achieving an impressive **92.7% accuracy** across 13 human activity classes.

EDUCATION

National Institute of Technology, Surat *Aug 2019 – May 2023*

*Bachelor of Technology in Computer Science and Engineering | CGPA: **9.04**, 7th rank in Department* Surat, India

PUBLICATIONS & PATENTS

- **Publication:** "Who's Related? Fast and Accurate Family Relationship Detection in Conversations," **ICASSP 2026**, Barcelona, Spain. **[Link]**
- **Publication:** "Screen2UI: Pixel-Only On-Device UI Hierarchy Generation," **ICVGIP 2025**. **[Link]**
- **Patent:** "System and Method for Adaptive User Interface Analysis and Dynamic Interactive Task Guidance"

ACHIEVEMENTS

- **MD Special Recognition Award (Samsung R&D Institute India – Bangalore):** Recognized for outstanding contribution to **Bixby 3.0 LLM-powered voice control** and advancements in **AI & NLP technologies**. *Feb 2025*
- **Samsung Excellence Award – Team Awesome Award:** Awarded for exceptional teamwork and contribution to the **OnDevice AI Team's innovation journey** at Samsung R&D Bangalore. *July 2024*
- **Samsung AI Expert (Workera):** Core-1: Mathematics for AI, Probability and Statistics, Prompt Engineering; Core-2: Machine Learning, Deep Learning, AI Fairness, PyTorch, TensorFlow; Core-3: Computer Vision; Core-4: Diffusion Models.
- **JEE Mains:** Achieved 99.28 percentile among 900,000 candidates, securing an All-India Rank of 8420.

TECHNICAL SKILLS

Machine Learning, Deep Learning & AI:

- Agentic AI (LangGraph, LangChain, CrewAI, OpenAI API, LlamaIndex, Retrieval-Augmented Generation (RAG), Pinecone, Vector DBs)
- NLP and Computer Vision (Neural Networks, CNN, LSTM, Transformers, BERT)
- Supervised & Unsupervised Learning (Regression, Decision Trees, Random Forests, SVM, Clustering: K-Means, XGBoost)
- Large Language Models (LLMs), Generative AI, Named Entity Recognition (NER)

Programming Languages & Frameworks/Libraries:

- Python, C, C++, SQL, TensorFlow, PyTorch, Keras, Hugging Face, Scikit-learn, OpenCV
- Agentic-AI (CrewAI, N8N), NumPy, Pandas, Matplotlib, Seaborn, NLTK, SpaCy

Big Data & Cloud Technologies:

- Google Cloud (Vertex AI), AWS (S3, SageMaker), Hadoop, Spark

Model Deployment & Optimization:

- On-Device AI, TensorRT, TFLite, ONNX, Flask, FastAPI, Docker, Kubernetes

Developer Tools:

- Git, GitHub, Linux, MySQL, MongoDB

PROJECTS

FinPilot Financial Advisor – Personalized Wealth Management through Multi-Agent AI

LangGraph, LangChain, Pinecone, RAG, OpenAI API, LlamaIndex **GitHub**

- Built a **multi-agent financial advisor AI system** with specialized **agents (Portfolio Analyzer, Finance Query Box, Watchlist Analyzer)** for intelligent investment insights, real-time monitoring, and personalized portfolio analysis.
- Developed a scalable **multi-agent orchestration framework** using **LangGraph, OpenAI API, RAG, Binance API, Tavily web search, and financial document parsing**, reducing manual financial data processing effort by 60%.

POSITIONS OF RESPONSIBILITY

Mentor & Relationship Manager *May 2024 – Nov 2024*

Samsung Research and Development | OnDevice AI Team

- Mentored 2-month and 6-month interns, providing end-to-end guidance on **AI/ML project design**, model development, and **deployment workflows**; facilitated best practices in coding, experimentation, and collaborative research within the Intelligent Platform Team.